Answers learning task 13

2026 – 2027

Exercise 8.4: A confidence interval estimate is desired for the gain in a circuit on a semiconductor device. Assume that gain is normally distributed with standard deviation $\sigma = 25$.

(a) Find a $95\,\%$ CI for μ when $n = 10$ and $\bar{x} = 1000$.

(b) Find a $95\,\%$ CI for μ when $n = 25$ and $\bar{x} = 1000$.

(c) Find a $99\,\%$ CI for μ when $n = 10$ and $\bar{x} = 1000$.

(d) Find a $99\,\%$ CI for μ when $n = 25$ and $\bar{x} = 1000$.

(e) How does the length of the CIs computed above change with the changes in sample size and confidence level?

Answer

Whenever we want a symmetric $100 \cdot (1 - \alpha)$ % confidence interval for the mean μ (in case σ is known), we use the formula

$$[\overline{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \overline{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}]$$

Here, if we want a $95\,\%$ confidence interval, that means $100 \cdot (1 - \alpha) = 95$, or $\alpha = 0.05$. The corresponding value of $z_{\alpha/2} = z_{0.025}$ can be found by taking the $1 - \alpha/2$-percentile of the standard normal distribution $N(\mu = 0, \sigma = 1)$:

$$z_{0.025} = \text{InvNorm}(\text{area} = 1 - \alpha/2 = 0.975, \mu = 0, \sigma = 1) \approx 1.9600.$$

Similarly, for $99\,\%$ confidence intervals ($\alpha = 0.01$), we find:

$$z_{0.005} = \text{InvNorm}(\text{area} = 1 - \alpha/2 = 0.995, \mu = 0, \sigma = 1) \approx 2.5758.$$

(a) Filling in $\overline{x} = 1000$, $\sigma = 25$, $n = 10$, $\alpha = 0.05$ gives $[984.5051, 1015.4949]$

(b) Filling in $\overline{x} = 1000$, $\sigma = 25$, $n = 25$, $\alpha = 0.05$ gives $[990.2002, 1009.7998]$

(c) Filling in $\overline{x} = 1000$, $\sigma = 25$, $n = 10$, $\alpha = 0.01$ gives $[979.6363, 1020.3637]$

(d) Filling in $\overline{x} = 1000$, $\sigma = 25$, $n = 25$, $\alpha = 0.01$ gives $[987.1209, 1012.8791]$

(e) Increasing the sample size n decreases the length of the CI, whereas increasing the confidence level increases the length of the CI.

Exercise 8.7: Consider the gain estimation problem in Exercise 8.4.

(a) How large must n be if the length of the $95\,\%$ CI is to be 30?

(b) How large must n be if the length of the $99\,\%$ CI is to be 30?

Answer

Notice that the length of a confidence interval can be computed by

$$\begin{aligned}\text{length} &= \text{upper bound} - \text{lower bound} \\ &= (\overline{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}) - (\overline{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}) \\ &= 2 \cdot z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}.\end{aligned}$$

If we want to constrain the length of a confidence interval to a value L, we need to bound this quantity from above:

$$2 \cdot z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq L \Rightarrow n \geq \left(\frac{z_{\alpha/2} \cdot \sigma}{\frac{1}{2}L} \right)^2.$$

(a) Filling in $z_{\alpha/2} = 1.9600$, $\sigma = 25$ and $L = 30$ gives $n \geq 11$ (notice that the sample size n is always an integer, so to satisfy the guarantee we need to round up to the nearest integer.)

(b) Filling in $z_{\alpha/2} = 2.5758$, $\sigma = 25$ and $L = 30$ gives $n \geq 19$

Exercise 8.9: Suppose that $n = 100$ random samples of water from a freshwater lake were taken and the calcium concentration (mg/L) measured. A $95\,\%$ CI on the mean calcium concentration is $0.52 \leq \mu \leq 0.74$.

(a) Would a $99\,\%$ CI calculated from the same sample data be longer or shorter?

(b) Consider the following statement: "There is a $95\,\%$ chance that μ is between 0.52 and 0.74." Is this statement correct? Explain your answer.

(c) Consider the following statement: "If $n = 100$ random samples of water from the lake were taken and the $95\,\%$ CI on μ computed, and this process were repeated 1000 times, 950 of the CIs would contain the true value of μ." Is this statement correct? Explain your answer.

Answer

(a) A confidence interval with a higher level of confidence always has to be longer if the other factors at play stay the same.

(b) No, we cannot speak of probabilities, as μ is a parameter and therefore has a deterministic value. The problem here is that people often confuse "a parameter with a unknown, but fixed value" with "a random variable". Correct interpretation: when a large number of samples is collected and a $95\,\%$-confidence interval for μ is constructed, independently for each of this samples, then the true mean μ will lie inside (approximately) $95\,\%$ of the computed intervals.

(c) Yes, see explanation part (b).

Exercise 8.11: The yield of a chemical process is being studied. From previous experience, yield is known to be normally distributed and $\sigma = 3$. The past five days of plant operation have resulted in the following percent yields: 91.6, 87.34, 90.8, 89.65, and 91.3. Find a $95\,\%$ two-sided confidence interval on the true mean yield.

Answer

This question requires the additional step that we first need to compute the sample mean of the sample:

$$\overline{x} = \frac{91.6 + 87.34 + 90.8 + 89.65 + 91.3}{5} = 90.138.$$

Filling in $\overline{x} = 90.138$, $\sigma = 3$, $n = 5$ and $\alpha = 0.05$ gives $[87.5084, 92.7676]$.

Exercise 8.17: The life in hours of a 75-watt light bulb is known to be normally distributed with $\sigma = 20$ hours. Suppose you wanted the total width of the two-sided confidence interval on mean life to be six hours at $95\,\%$ confidence. What sample size should be used?

Answer

Filling in $z_{\alpha/2} = 1.9600$, $\sigma = 20$ and $L = 6$ gives $n \geq 171$ (notice that the sample size n is always an integer, so to satisfy the guarantee we need to round up to the nearest integer).

Exercise 8.29: A postmix beverage machine is adjusted to release a certain amount of syrup into a chamber where it is mixed with carbonated water. A random sample of 25 beverages was found to have a mean syrup content of $\bar{x} = 30$ mL. Suppose the standard deviation is known and equal to $\sigma = 0.5$ mL. Find a $95\,\%$ CI on the mean volume of syrup dispensed.

Answer

Filling in $\overline{x} = 30$, $\sigma = 0.5$, $n = 25$, $\alpha = 0.05$ gives $[29.8040, 30.1960]$.

Exercise 7.63: Let $f(x) = (1/\theta)x^{(1-\theta)/\theta}$, $0 < x < 1$, and $0 < \theta < \infty$. Show that $\hat{\Theta} = -(1/n)\sum_{i=1}^{n} \ln(X_i)$ is the maximum likelihood estimator for θ and that $\hat{\Theta}$ is an unbiased estimator for θ.

Answer

The likelihood function is $L(\theta) = \theta^{-n} \prod_{i=1}^{n} x_i^{(1-\theta)/\theta}$. Taking the natural log gives:

$$\ln L(\theta) = -n \ln(\theta) + \frac{1-\theta}{\theta} \sum_{i=1}^{n} \ln(x_i)$$

Setting the derivative with respect to θ equal to zero:

$$\frac{\mathrm{d}\left(\ln L(\theta)\right)}{\mathrm{d}\theta} = -\frac{n}{\theta} - \frac{1}{\theta^2} \sum_{i=1}^{n} \ln(x_i) = 0 \implies \hat{\Theta} = -\frac{1}{n} \sum_{i=1}^{n} \ln(X_i)$$

Next, we compute the expectation of the natural log of X_i: for this, we need to perform partial integration, with $f(x) = -\ln x \Rightarrow f'(x) = -\frac{1}{x}$ and $g'(x) = x^{\frac{1-\theta}{\theta}} \Rightarrow g(x) = x^{\frac{1}{\theta}}$.

$$\begin{aligned} E(-\ln X_i) &= \int_{x=0}^{1} \ln x \cdot \frac{1}{\theta} x^{\frac{1-\theta}{\theta}} \,\mathrm{d}x \\ &= \left[-\ln x \cdot x^{\frac{1}{\theta}} \right|_{x=0}^{1} - \int_{x=0}^{1} x^{\frac{1-\theta}{\theta}} \,\mathrm{d}x \\ &= (0 - 0) - \left[\theta x^{\frac{1}{\theta}} \right|_{x=0}^{1} \\ &= \theta. \end{aligned}$$

We can use this result to show that $\hat{\Theta}$ is indeed an unbiased estimator:

$$E(\hat{\Theta}) = E\left(\frac{1}{n}\sum_{i=1}^{n} -\ln X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E\left(-\ln X_i\right) = \frac{1}{n}\cdot(\theta+\theta+\ldots+\theta) = \theta.$$

Hence, $\hat{\Theta}$ is an unbiased estimator for θ.